INCH-POUND

MIL-F-50864B (AR)
AMENDMENT 4
29 August 2000
SUPERSEDING
AMENDMENT 3
31 July 1993

MILITARY SPECIFICATION

FUZE, PIBD, M549, LESS SPITBACK
METAL PARTS AND
LOADING, ASSEMBLING AND PACKAGING

This amendment forms a part of MIL-F-50864B (AR), dated 7 March 1988, and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Add under Military Standards:

"MIL-STD-414	-	Sampling procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-1168	-	Ammunition Lot Numbering"

* 2.1.2, Add the following:

"8886327	-	Cartridge Case Assembly 40MM M169
9287854	-	Body Assembly
9399407	-	Capsule Assembly
12977504	-	Body Insert
12977505	-	Test Vehicle for M549/M549A1 Fuze
MS28900-28	-	Packing, Preformed, for Electrical Use

2.1.2, Delete "9349971 …" and substitute:

"9349971	-	Cartridge, 40MM, Test Vehicle with M549 Fuze"

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Add under 2.2:

"ASTM C 208 - Insulating Board (Cellulosic Fiber), Structural and Decorative"

3.3, Delete "Safety spring not engaged in teeth of rotor gear" and substitute "Safety spring not locking rotor and gear assembly".

3.4, Delete "… either the air pressure, functioning…" and substitute "… either the functioning…".

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4.3.1, Add:

"Ogive, housing escapement with crimped bottom plate	9287861		50"

Table I, Delete:

"Arming time, minimum	125	3.2	4.5.12"

substitute:

"Arming time, minimum	125 1/	3.2	4.5.12"

Notes: Add "1/ The one hundred and twenty-five samples shall be tested a total of five times each without failure."

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Table I, Delete:

"Air pressure	125	3.2	4.5.20"

Add:

"Ogive, housing escapement with crimped bottom plate (Dwg. 9287861) Torque test	50	3.2	4.5.20"

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Change from “1/2 or 5/8 inch Celotex” to “1/2 or 5/8 inch thick insulating board”

PAGE 12

4.4.2.1, Delete Minor 201 in its entirety.

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4.4.2.4, Major 103, Under No. of Sample Units column, delete “4”.

Delete Note 1 in its entirety and substitute:

“1/ Sampling and rejection shall be in accordance with MIL-C-5541 or MIL-A-8625 as applicable.”

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4.4.2.7, Delete “Major 101” in its entirety.

PAGE 20

4.4.2.9, Delete Major 106 from “Diameter across legs” and substitute “Diameter across legs, min”.

Delete: Major 106 and substitute:

“106. Leg broken, bent or missing 0.40% 3.2 Visual”

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4.4.2.16, Delete 101 and substitute:

“101. Push test of actuator 125 1/ Acc-1 1/ Rej-2 3.2 4.5.10”

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Add: Under Notes:

"1/ For ogive assembly lots of 22,000 or less, use MIL-STD-414. Inspection level II, Single Specification Limit. Variability unknown. Range Method. AQL = .40%. If the calculated value of (X-L)/R is less than "k" or if any individual measurement is less than 400 pounds, the lot shall be rejected."

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4.4.2.18: Delete "safety spring…sector gear" Critical 4 and substitute:

"Safety spring end not locking rotor and gear assembly 5/"

Delete: Critical 5 in its entirety and substitute:

"5. Drive Screw missing 100% 3.2 AIE/4.5.14"

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4.4.2.21: Add:

"CRITICAL

6. X-ray examination of fuze assembly 100% 3.3 4.5.16.1"

* 4.4.2.21, Critical 6, Delete in its entirety.

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4.4.2.21: Delete Major 101 in its entirety and substitute:

"101. Torque Test 5 per shift 2/ 0-1 3.2 4.5.20"

Major 102, 103 and 104, delete "100 1/ 2/" and substitute 100 1/"

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Delete Note 2/ in its entirety and substitute the following:

"2/ The 5 samples shall be by assembling ogive, escapement housing and bottom plate only using normal assembly procedure. If any sample fails the drawing requirement, the shift's production represented by the sample shall be rejected."

4.4.2.21, Major 102, No. of Sample Units column, delete "100 1/" and substitute "100 1/ 2/ 4/"

Add under Notes:

"4/ Acceptance is based on requirements of paragraph 3.7 or disassembly of the round and examination of the M59 detonator for functioning in full armed position."

Major 103, delete "1/2 or 5/8 inch Celotex" and substitute "1/2 or 5/8 inch thick insulating board"

Delete: Major 111 in its entirety and substitute:

"111. Perpendicularity of large outside diameter with crimped surface 0.40% 3.2 Gage"

Add Paragraph 4.5.16.1:

"4.5.16.1 X-Ray examination of fuze assembly. The fuze assembly shall be x-rayed using approved automatic test equipment and test techniques (see 6.4). The equipment shall be in accordance with Dwg. 12903079. Inspection shall be conducted in two (2) planes: from the top of the fuze assembly down; and from the side. The side view shall show the setback pin adjacent to the rotor. Observance shall be made for the following critical and major defects:

a. Setback pin not adjacent to the rotor gear.
b. Setback pin not seated in bottom of counterbore of escapement housing.
c. Setback pin missing.
d. Setback spring missing.
e. Safety spring missing

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f. Safety spring end not in contact with root diameter surface or seated within root between first 2 teeth of rotor gear.
g. O-ring missing.
h. Pinion assembly missing.
i. Verge assembly missing.
j. Rotor gear assembly in armed or partially armed position.
k. Drive screw missing or not fully seated.
l. A major defect is defined as any defect not specified as critical above."

* 4.5.16.1, delete in its entirety.

4.5.17: Delete in its entirety and substitute the following:

"4.5.17 The fuze assemblies packed in their regular shipping containers shall be tested in accordance with test B1, Section 6.1.2 of MIL-STD-331. The test shall be performed at ambient temperature only. After completion of the test, the fuze assemblies shall be x-rayed as specified in 4.5.16. Observation shall be made for compliance with the requirements specified in 3.4."

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4.5.20: Delete in its entirety and substitute the following:

"4.5.20 Torque test. Sample shall be placed in an approved fixture and the torque specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This test is a destructive test. Parts so tested shall not be returned to the lot."

* 4.5.21, Add the following at the end of the paragraph:

"Approved alternate configuration (12977505, Test Vehicle for M549/M549A1 Fuze). Assemble capsule assembly, 9399407 into body, inert, 12977504 and this assembly into body assembly, 9287854. Assemble Packing, MS28900-28, and cartridge case, 8886327 to body assembly. Thread the fuze,

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PIBD, M549, less spitback, 9287861, onto the body assembly. Verify that there is a .005 inch minimum gap between the projectile body and the shoulder of the fuze."

4.5.21.2: Delete "One-half (1/2) or five-eights (5/8) inch Celotex" and substitute "One-half (1/2) or five-eighths (5/8) inch insulating board" and "1/2 or 5/8 inch piece of Celotex." and substitute "1/2 or 5/8 inch insulating board per ASTM C 208 or approved equal."

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4.5.22: Delete "1/2 or 5/8 inch piece of Celotex" and substitute "1/2 or 5/8 inch insulating board per ASTM C 208 or approved equal."

* 4.5.22, line 2, after "Transportation vibration test…" Add:

"The approved alternate configuration (see 4.5.21) shall be used, or…"

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